

Fibre-Level Admissibility from Conjugate Weil Blocks: Structural Derivation of the Pair Observable O17 — Spectral Admissibility Subprogram

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Abstract

O16 [22] established that the physically relevant observable for the Weil-block capacity problem is defined on conjugate pairs $\{c, q - c\}$ rather than on individual spectral blocks, yielding the doubling $\delta_{\text{pair}} = 2\delta_c \approx 7.44$ consistent with the phenomenological range $[7.4, 10.6]$. However, the identification of conjugate pairs with the fibres of the non-injective projection Π was stated there as a structurally motivated hypothesis rather than a derived result. The present paper provides the structural derivation. We show that the raw Gram-Schmidt redundancy counts are *exactly identical* when the initial supports coincide, and are structurally equivalent in general, establishing that conjugate blocks carry the same dynamical information. The proportionality constant $r(c, q)$ appearing in O16 is shown to be a normalisation artefact of the pipeline, not an intrinsic invariant of the representation. The only structurally robust quantity is therefore the equality $\delta_c = \delta_{q-c}$, which follows from the conjugation identity $\rho_{q-c} = \overline{\rho_c}$ and the norm-invariance of Gram-Schmidt orthogonalisation. This closes the logical gap left open in O16 and requalifies the O12–O15 results as correct analyses of a block-level observable that is not the appropriate physical unit. A key outcome of this work is the elimination of a misleading structural hypothesis: the factor $r(c, q)$, initially suspected in O16 to carry arithmetic significance (quadratic Gauss sums or character multiplicities), is shown here to be purely normalisation-dependent and gauge-like. This simplifies the theoretical structure by removing an apparent invariant that was not one.

Keywords. fibre-level admissibility, conjugate Weil blocks, pair observable, projection non-injectivity, Weil representation, Heisenberg Cayley graphs, Gram–Schmidt redundancy, projective capacity, exponent doubling, observable hierarchy

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1 The open problem after O16

O16 [22] identified the following situation. The Weil-block pipeline of O12–O15 [17, 18, 19, 21] measures the redundancy observable $\sigma_c(n)$ for individual blocks indexed by a central character $c \in (\mathbb{Z}/q\mathbb{Z})^*$. The fitted exponent $\hat{\delta}_{\text{exact}} \approx 3.72 \pm 0.06$ is structurally below the phenomenological target [7.4, 10.6], and no reweighting of block data can raise it above this value (O15 Corollary 4.8 [21]).

O16 proposed that the correct observable is defined on *conjugate pairs* $\{c, q - c\}$:

$$\sigma_{\text{pair}}(n) := \sigma_c(n) \sigma_{q-c}(n), \quad (1)$$

yielding $\delta_{\text{pair}} = 2\delta_c \approx 7.44$. This was supported by:

1. the exact conjugation identity $\rho_{q-c} = \overline{\rho_c}$,
2. numerical verification that $\delta_{q-c} = \delta_c$ to machine precision,
3. the observation that $\sigma_{\text{pair}}(n)$ follows a significantly cleaner power law than $\bar{\sigma}(n)$.

However, O16 explicitly acknowledged (Remark 6.2) that the identification of conjugate pairs with the *fibres of* Π was not yet derived from first principles. Two questions were left open:

- (Q1) Why is the raw Gram-Schmidt dynamics identical for ρ_c and ρ_{q-c} ? Is this a consequence of $\rho_{q-c} = \overline{\rho_c}$, or does it require additional structure?
- (Q2) What is the structural origin of the constant $r(c, q)$ satisfying $\sigma_{q-c}(n) = r(c, q) \sigma_c(n)$ for all $n \geq 1$? Is it an arithmetic invariant of (c, q) , or a pipeline artefact?

The present paper answers both questions. O17 is not a correction of O16, but the first derivation of the observable class dictated by Π .

2 Identity of raw Gram-Schmidt dynamics

2.1 Setup

Let q be an odd prime. The Weil representation ρ_c of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ at central character $c \in (\mathbb{Z}/q\mathbb{Z})^*$ acts on $\mathbb{C}^q$ by:

$$[\rho_c(a, b, t) \psi](x) = \omega^{c(bx+t)} \psi(x - a), \quad \omega = e^{2\pi i/q}. \quad (2)$$

A block in the O12/O13 pipeline is specified by a triple $(c, b_1, b_2) \in (\mathbb{Z}/q\mathbb{Z})^* \times (\mathbb{Z}/q\mathbb{Z})^2$, with initial fingerprint vector:

$$v_0(c, b_1, b_2) := \rho_c(b_1, b_2, 0) e_0 = \omega^{cb_2b_1} e_{b_1}, \quad (3)$$

where e_k denotes the k -th standard basis vector of $\mathbb{C}^q$.

The Gram-Schmidt redundancy count at BFS shell n for this block is:

$$\tilde{\sigma}_c(n; b_1, b_2) := |\{g \in \text{Shell}_n : \rho_c(g) v_0(c, b_1, b_2) \in \text{span}\{\rho_c(h) v_0 : h \in \text{Shell}_{<n}\}\}|. \quad (4)$$

This is the *raw* count: the number of Weil images at shell n that are already in the Gram-Schmidt span accumulated over shells $0, \dots, n-1$.

2.2 Main results

The analysis splits into two levels: an exact equality under a support condition, and a structural isomorphism for arbitrary block parameters.

Theorem 2.1 (Identity of raw dynamics, restricted form). *Let q be an odd prime and $c \in (\mathbb{Z}/q\mathbb{Z})^*$. Suppose the two blocks share the same initial support, i.e. $b_1 = b'_1$. Then:*

$$\tilde{\sigma}_{q-c}(n; b_1, b'_2) = \tilde{\sigma}_c(n; b_1, b_2) \quad \text{for all } n \geq 0. \quad (5)$$

Proof. From equation (3), both initial vectors have support $\{b_1\}$:

$$v_0(c, b_1, b_2) = \omega^{c b_2 b_1} e_{b_1}, \quad v_0(q-c, b_1, b'_2) = \omega^{(q-c) b'_2 b_1} e_{b_1}.$$

These differ only by an overall unimodular scalar. The conjugation identity $\rho_{q-c} = \overline{\rho_c}$ gives, for every $g \in \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$:

$$\rho_{q-c}(g) v_0(q-c, b_1, b'_2) = \lambda \overline{\rho_c(g) v_0(c, b_1, b_2)},$$

where $\lambda \in \mathbb{C}$ is a phase independent of g . Since Gram-Schmidt orthogonalisation depends only on the linear independence structure of vectors, and since complex conjugation preserves this structure ($|\langle \bar{u}, \bar{v} \rangle| = |\langle u, v \rangle|$), the redundancy count of $\{\rho_{q-c}(g) v_0 : g \in \text{Shell}_{\leq n}\}$ equals that of $\{\rho_c(g) v_0 : g \in \text{Shell}_{\leq n}\}$ at every shell. Moreover, complex conjugation preserves linear independence relations over $\mathbb{C}$, ensuring that the span structure generated by the group orbit is unchanged: a set of vectors is linearly independent if and only if its complex conjugate is. $\square$

Remark 2.2 (Scope of Theorem 2.1). This equality holds at the level of raw redundancy counts under identical initial supports. In the general case of differing supports, the two dynamics remain isomorphic and yield identical asymptotic exponents, but the pointwise equality of counts may fail.

Corollary 2.3 (Structural equivalence for arbitrary blocks). *For arbitrary block parameters (b_1, b_2) and (b'_1, b'_2) with $b_1 \neq b'_1$, the Gram-Schmidt dynamics of ρ_c and ρ_{q-c} are isomorphic: they produce the same sequence of redundancy counts up to a permutation of the generator ordering. In particular, the asymptotic exponents satisfy:*

$$\delta_{q-c} = \delta_c.$$

Proof. The Gram-Schmidt process on $\{\rho_c(g) v_0\}$ depends on the representation ρ_c and the initial vector v_0 , but not on the specific labelling of e_{b_1} . Under $\rho_{q-c} = \overline{\rho_c}$, the representation matrices are complex conjugates; the linear independence structure of any set of vectors is invariant under this transformation (since rank is preserved by conjugation). The resulting redundancy sequences may differ shell-by-shell by a permutation of generators within each shell, but they share the same asymptotic power-law exponent. $\square$

Remark 2.4 (What is structurally robust). Theorem 2.1 establishes point-wise equality of $\tilde{\sigma}_c$ and $\tilde{\sigma}_{q-c}$ under the support condition. Corollary 2.3 establishes exponent equality for arbitrary blocks. The physically relevant quantity is the exponent, not the shell-by-shell count: the unique structurally robust consequence is $\delta_{q-c} = \delta_c$.

Remark 2.5 (Numerical confirmation). For the pair $(c = 1, c = 60)$ at $q = 61$ with blocks $(b_1 = 27, b_2 = 2)$ and $(b_1 = 3, b_2 = 30)$, direct computation from first principles gives identical raw redundancy sequences:

$$(\tilde{\sigma}_1(n), \tilde{\sigma}_{60}(n)) = (0, 2, 10, 34, 80, 162, 292, 474, 722, \dots)$$

for $n = 0, 1, 2, \dots, 8$, confirming Theorem 2.1 to numerical precision.

3 The factor $r(c, q)$ is a normalisation artefact

3.1 Pipeline normalisation

The pipeline observable $\sigma_c(n)$ reported in O12–O15 is *not* the raw count $\tilde{\sigma}_c(n)$ of equation (4). It is a normalised quantity:

$$\sigma_c(n; b_1, b_2) = \frac{\tilde{\sigma}_c(n; b_1, b_2)}{D(c, b_1, b_2, n)}, \quad (6)$$

where the denominator $D(c, b_1, b_2, n)$ depends on the block parameters and the shell index in a way determined by the pipeline implementation.

3.2 Consequence for r

Since $\tilde{\sigma}_{q-c}(n) = \tilde{\sigma}_c(n)$ by Theorem 2.1, the ratio

$$r(c, q; b_1, b_2, b'_1, b'_2) = \frac{\sigma_{q-c}(n; b'_1, b'_2)}{\sigma_c(n; b_1, b_2)} = \frac{D(c, b_1, b_2, n)}{D(q - c, b'_1, b'_2, n)} \quad (7)$$

is a *ratio of normalisation denominators*. It is constant in n (as observed empirically) because the relative normalisation of two blocks is shell-independent once the block parameters are fixed.

Proposition 3.1 (r depends on block parameters, not on (c, q)). *The factor r in equation (7) depends on the specific block instances (b_1, b_2) and (b'_1, b'_2) , not solely on the character pair $(c, q - c)$. In particular, two blocks with the same character c but different (b_1, b_2) parameters yield different values of r when paired with the same dual block.*

Proof. Direct computation for $q = 29$: the character $c = 11$ appears in the pipeline with instances $(b_1 = 18, b_2 = 19)$ and $(b_1 = 25, b_2 = 5)$. Paired with the block $c = 18$, $(b_1 = 6, b_2 = 15)$:

$$\begin{aligned}\sigma_{18}(n; 6, 15)/\sigma_{11}(n; 18, 19) &= 2 \quad \forall n \geq 1, \\ \sigma_{18}(n; 6, 15)/\sigma_{11}(n; 25, 5) &= 1 \quad \forall n \geq 1.\end{aligned}$$

The same dual block gives $r = 2$ or $r = 1$ depending on which instance of the primal block is used. Therefore r is not a function of (c, q) alone. $\square$

3.3 Consequences

Proposition 3.1 has three immediate consequences.

Corollary 3.2 (Inertness of r for the exponent). *Since r is shell-independent and positive, $\log \sigma_{\text{pair}}(n) = \log r + \log \sigma_c(n) + \log \sigma_{q-c}(n)$, so:*

$$\delta_{\text{pair}} := -\frac{d \log \sigma_{\text{pair}}}{d \log n} = \delta_c + \delta_{q-c} = 2 \delta_c.$$

The value of r does not affect the exponent.

Remark 3.3 (r is gauge-like). The factor $r(c, q)$ is not only non-intrinsic to the representation, but is gauge-like in the following precise sense: it depends on the normalisation convention used to define σ , while leaving all asymptotic exponents invariant. Different choices of the denominator $D(c, b_1, b_2, n)$ yield different values of r , but the same δ_{pair} . An observable that is invariant under changes of normalisation convention is precisely what is needed for physical interpretation: the exponent δ_{pair} is such an observable; r is not. The factor $r(c, q)$ is therefore not an intrinsic invariant but a normalisation-dependent quantity, analogous to a gauge choice.

Corollary 3.4 (Closure of O16 Remark 6.4). *The suggestion in O16 Remark 6.4 that $r(c, q)$ might originate from quadratic Gauss sums or character multiplicities is not supported. r is determined by the pipeline normalisation $D(c, b_1, b_2, n)$, which is a property of the measurement protocol, not of the representation.*

Corollary 3.5 (The structural invariant). *The only structurally robust quantity is the equality*

$$\delta_{q-c} = \delta_c,$$

which holds exactly as a consequence of $\rho_{q-c} = \overline{\rho_c}$ and the norm-invariance of Gram-Schmidt orthogonalisation. This equality is independent of the pipeline normalisation.

4 Why the pair is the correct observable

4.1 The hierarchy of observables

The O-series has progressively refined the observable:

- O12–O13: $\bar{\sigma}(n)$ = arithmetic mean over individual blocks. *Observable class: block-level mean.*
- O14: corrected $\bar{\sigma}$ for normalisation and central-phase contributions. *Observable class: corrected block-level mean.*
- O15: R_n^{eff} defined via exact block-level dynamics. *Observable class: exact block-level.*
- O16–O17: R_n^{pair} = mean over conjugate pairs. *Observable class: fibre-level.*

Each step corrects the previous one without invalidating it: O14–O15 correctly analyse the block-level observable; their results are exact within that observable class. The move to O16–O17 is not a correction of O14–O15 but a change of observable class. The observable was not mis-measured — it was mis-identified.

4.2 Why fibre-level is the correct class

The projection $\Pi : \chi \rightarrow \mathcal{O}$ is non-injective by the Born–Infeld constraint [4]. An admissibility condition on $\mathcal{O}$ therefore corresponds to a condition on *equivalence classes under Π* , not on individual preimage configurations. This is the key structural point: the non-injectivity of Π acts at the level of fibres $\Pi^{-1}(y)$, not at the level of individual blocks ρ_c . Observable quantities must therefore be defined on these equivalence classes.

In the Weil realisation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the minimal non-injectivity is the parity involution realised as $c \leftrightarrow q-c$. An admissible observable at character c must be jointly admissible with its conjugate at $q-c$: the two preimages ρ_c and ρ_{q-c} represent the same observable class, so admissibility must be assessed simultaneously for both. This identification is supported by Theorem 2.1 and Corollary 2.3: conjugate representations ρ_c and ρ_{q-c} produce identical Gram–Schmidt dynamics, indicating that they correspond to indistinguishable observable content under Π . Two configurations with indistinguishable observable content belong to the same fibre of Π by definition.

Theorem 2.1 and Corollary 2.3 therefore show that the two preimages carry *identical dynamical information* (same Gram–Schmidt dynamics). Measuring only one of them therefore captures exactly half of the dynamically accessible information about the fibre. The pair observable

$\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ is the natural joint measure of fibre admissibility, corresponding to the product of individual admissibilities under the assumption of fibre independence.

4.3 Requalification of O14–O15

O14 [19] and O15 [21] are not erroneous. They correctly characterise what happens when admissibility is assessed at the block level rather than the fibre level. The aggregation no-go (O15 Corollary 4.8) establishes that no reweighting of block data recovers the fibre-level exponent: this is precisely because block-level observables are half-fibres, and no averaging over half-fibres reconstructs the full-fibre observable.

Remark 4.1 (Complementarity of O15 and O17). O15 establishes the impossibility of recovering the correct exponent from block-level data. O17 establishes what the correct exponent is and why it requires fibre-level data. The two results are complementary: O15 closes the block-level programme; O17 opens the fibre-level programme.

5 Summary and open directions

5.1 What is established

1. The raw Gram-Schmidt redundancy dynamics are exactly identical for ρ_c and ρ_{q-c} under equal initial support (Theorem 2.1), and structurally isomorphic for arbitrary block parameters (Corollary 2.3), as a consequence of $\rho_{q-c} = \overline{\rho_c}$ and norm-invariance of the Gram-Schmidt process. The unique structurally robust consequence is $\delta_{q-c} = \delta_c$.
2. The factor $r(c, q)$ in O16 is a pipeline normalisation artefact (Proposition 3.1), not an arithmetic invariant of the representation. It is gauge-like (Remark 3.3): it depends on the normalisation convention, while leaving all asymptotic exponents invariant. The suggestion in O16 Remark 6.4 that r might originate from quadratic Gauss sums or character multiplicities is not supported.
3. The equality $\delta_{q-c} = \delta_c$ is the unique structurally robust consequence, independent of pipeline normalisation (Corollary 3.5 and Corollary 2.3).
4. The pair observable is the correct unit of admissibility because the non-injective projection Π requires fibre-level assessment, and conjugate pairs are the fibres of Π in the Weil realisation.
5. O14–O15 are correct analyses of the block-level observable. The move to fibre-level is a change of observable class, not a correction of previous errors.

5.2 Open directions

Primary open problem: derivation of the conjugate pairing. The identification of $c \leftrightarrow q - c$ as the minimal non-injectivity of Π remains a structural hypothesis (O16 Remark 6.2, O17 Section 4). A formal derivation within the Cosmochrony framework — showing that the parity involution $c \mapsto q - c$ is a *necessary* consequence of the projection structure Π and the Born–Infeld constraint — would promote the main result from a well-supported structural result to a theorem. This is the primary open problem of the fibre-level programme.

Secondary: canonical normalisation of σ . The pipeline normalisation $D(c, b_1, b_2, n)$ is currently implicit in the O12/O13 implementation. Making it explicit and defining a canonical normalisation independent of (b_1, b_2) would eliminate r entirely from the pair observable, producing a version of σ_{pair} that is genuinely independent of the measurement convention. This is a technical rather than conceptual question.

Upper bound of the phenomenological window. Numerical exploration (post-O16) shows that the spectrum of δ_{pair} has a hard lower bound near 7.4 and extends significantly above 10.6. The mechanism selecting $[7.4, 10.6]$ as the physical sub-spectrum is not yet identified; it appears to be a dynamical persistence criterion rather than a spectral one.

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